

System immutability: All configurations in the tool should be made in code (infra as code) such that the complete setup is immutable. No script should be run outside this code at any stage.

Complete automation: Build, test, validate and deploy should be automated as much as possible, to avoid human intervention and errors.

Flexible with respect to disposability: Everything in the cloud setup is transient and shortlived. Tools must enable regular updates to avoid unnecessary failure of services.

External configuration: Configurations like passwords, file location/path and credentials that can be changed should be stored externally from the image/binary to make life easier.

Constant telemetry: The tool must enable logs, event streams, audits, and alerts with visualisation dashboards so that any functional, operational and security issues can be monitored upfront.

Delegated governance: Business agility must be enabled by defining

SLAs for managed services, along with strict governance in the DevSecOps life cycle.

Life cycle management: Life cycle management should be independent of decoupling activities like upgrading, scaling and deploying services.

Some common native tools for cloud management are discussed below. Figure 4 highlights multi-cloud management tools.

AWS Outpost is suitable for on-premise usage in smaller organisations that have less hardware footprints, as it does not support custom hardware handling and server provisioning. Anthos and Arc, on the other hand, work relatively in the same way by using the Kubernetes engine to handle multi-cloud, on-premise or private cloud seamlessly. Kubernetes is an integral part of Anthos, whereas Azure Arc can work with Kubernetes and edge computing services.

AWS has improved Outpost by making EKS run on Fargate for serverless Kubernetes solutions to support hybrid cloud and multi-cloud

solutions. Anthos, Outpost and Arc are strong competitors in the hybrid cloud space.

In cloud management, it is good to have a platform that operates as a ‘single pane of glass’ for discovery and tracking of cloud assets, optimising cloud utilisation and controlling cloud services. Though there are many tools like Apptio Cloudability, Opsani and Matilda that offer this, ManageIQ is an open source community development platform offered and sponsored initially by the Red Hat Foundation.

ManageIQ started off as a native platform for the Red Hat OpenShift environment, both for private and public clouds, and later started supporting cloud service providers like Azure, AWS and GCP. It is agentless and can be installed easily in private cloud environments using Vagrant or Docker, as well as on a public cloud as a PaaS service using Docker or a virtual appliance image. Figure 5 shows the ManageIQ architecture.

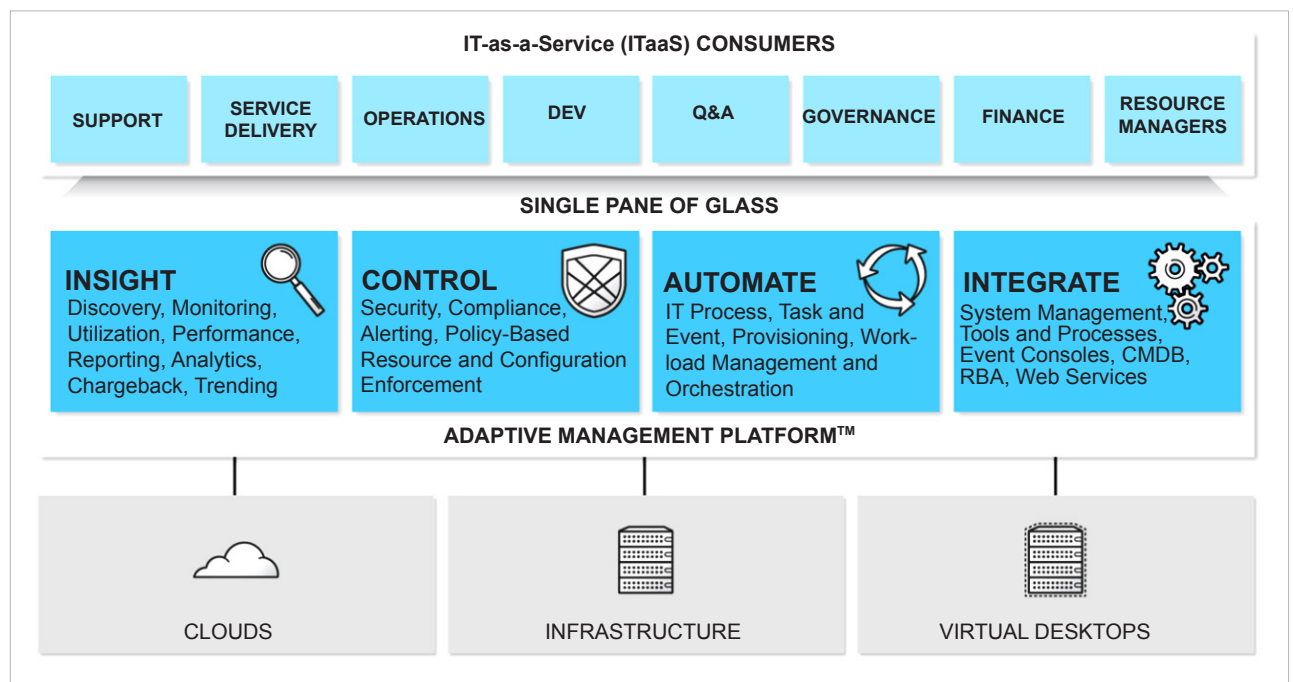


Figure 5: ManageIQ architecture (Image source: ManageIQ documentation)